FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PRINCIPLE OF STRUCTURE PROGRAMMING(PROG103)

ASSIGNMENT TITTLE : STRUCTURED DIGITAL SOLUTION FOR PUBLIC SERVICE (GUI-BASED APPLICATION)

Due Date : Week 11
NAME OF LECTURER : Mr. Alusine Jalloh
CLASS : BBIT1201F
SEMESTER/YEAR: SEMESTER 2/ YEAR 1

Academic Honesty Policy Statement

We, hereby attest that contents of this attachment are my own own work. Referenced works, articles, art, programs, papers or parts thereof are acknowledged at the end of this paper. This includes data excerpted from CD-ROMs, the internet, other private networks, and other people's disk of the computer system.

NO	NAMES	ID NUMBER	SIGNATURE
1	Abdul Tom Ngebeh	905005765	
2	Sarah Finda Squire	905005550	
3	Mariam Dolly Mans-Khanu	905006011	

Date:

for office use only upon

MENTOR’S COMMENTS/GRADE:

receiving

Remark

DATE :

TIME :

RECEIVER’S NAME:

Table of Contents

INTRODUCTION

The advancement of Information and Communication Technology (ICT) has significantly transformed the way organizations and businesses manage their daily operations. Modern businesses increasingly rely on computerized systems to improve efficiency, accuracy, and productivity in order to remain competitive in a rapidly changing business environment. The integration of technology into business processes has enabled organizations to automate routine tasks, reduce operational costs, improve decision-making, and enhance overall performance.

Small and Medium-Sized Enterprises (SMEs) play a vital role in the economic development of Sierra Leone through job creation, income generation, and the provision of goods and services. However, many SMEs continue to depend on traditional methods of record management, including notebooks, paper receipts, manual ledgers, and spreadsheets. These methods often result in challenges such as inaccurate record keeping, poor inventory management, delayed reporting, difficulty in tracking customer transactions, and inefficient financial management.

To address these challenges, the Smart Business Management System (SBMS) was developed as a desktop-based application that provides an integrated platform for managing various business operations. The system was developed using Python, Tkinter, and SQLite, and incorporates functionalities such as inventory management, customer management, vendor management, order processing, payment tracking, expense recording, and report generation. By centralizing business information and automating routine processes, the system improves operational efficiency, enhances data accuracy, and supports informed decision-making.

The Smart Business Management System serves as a practical solution for businesses seeking to transition from manual record-keeping methods to digital management systems. It demonstrates how software applications can be used to solve real-world business problems while promoting the adoption of technology in business operations. The project further highlights the application of structured programming principles and database management concepts in the development of a functional and user-friendly business management solution.

Problem Description

Small and Medium-Sized Enterprises (SMEs) play a significant role in the economic development of Sierra Leone by creating employment opportunities, generating income, and contributing to national growth. Despite their importance, many of these businesses still rely heavily on traditional methods of managing their daily operations. Business information such as product inventories, customer records, sales transactions, vendor details, payments, and expenses are often recorded manually using notebooks, paper receipts, ledgers, and, in some cases, basic spreadsheet applications. While these methods may appear convenient and affordable for small businesses, they become increasingly ineffective as the volume of business activities grows.

One of the major challenges associated with manual record management is the high risk of human error. Business owners and employees may make mistakes when recording transactions, calculating stock balances, or preparing financial summaries. Such errors can lead to inaccurate business records, financial losses, and poor decision-making. Additionally, manual systems make it difficult to monitor inventory levels effectively. Businesses may unknowingly run out of stock or overstock products due to inaccurate inventory records, resulting in reduced customer satisfaction and unnecessary expenses.

Another significant problem is the lack of proper customer and vendor management. In many businesses, customer information and supplier records are stored in separate notebooks or scattered documents. Retrieving this information can be time-consuming and inconvenient, especially when dealing with a large number of transactions. This situation often affects customer service quality and weakens supplier relationships.

Financial management also presents a major challenge. Many business owners struggle to accurately track payments, expenses, and profits. Without a centralized system, generating financial reports becomes difficult and time-consuming. Consequently, business managers often lack the information needed to evaluate business performance, identify profitable products, and make strategic decisions for growth and sustainability.

Furthermore, the absence of automated reporting mechanisms means that business owners spend considerable time compiling sales and expense information manually. This process not only reduces productivity but also increases the likelihood of errors in business reports. As competition increases and business environments become more dynamic, there is a growing need for digital solutions that can simplify business operations and improve efficiency.

The Smart Business Management System was therefore developed to address these challenges by providing a computerized platform that automates business processes, centralizes information storage, improves data accuracy, and supports effective decision-making. The system seeks to eliminate the limitations of manual record-keeping and provide businesses with a reliable tool for managing their operations efficiently.

System Explanation

The Smart Business Management System (SBMS) is a desktop-based business automation solution designed to simplify and improve the management of essential business activities. The system was developed using Python as the programming language, Tkinter as the graphical user interface framework, and SQLite as the database management system. These technologies were selected because they are reliable, efficient, and suitable for developing cost-effective business applications.

The system operates through a user-friendly graphical interface that allows users to interact with the application without requiring advanced technical knowledge. Through various forms and windows, users can enter, update, retrieve, and analyze business information. All data entered into the system is stored securely within a centralized SQLite database, ensuring consistency, reliability, and easy retrieval of records whenever required.

The operation of the system begins with the login module, which provides authentication and access control mechanisms. This module ensures that only authorized users can access business information, thereby improving security and protecting sensitive records from unauthorized access.

After successful authentication, users are directed to the dashboard, which serves as the central control panel of the application. The dashboard provides an overview of key business activities and presents important statistics such as the number of products, customers, vendors, orders, sales, expenses, and profits. This enables management to quickly assess the overall performance of the business.

The product management component allows businesses to maintain detailed information about products. Users can add new products, update existing records, delete obsolete items, and search for products when needed. The system automatically tracks inventory levels and generates warnings whenever stock falls below a predefined threshold. This functionality helps prevent stock shortages and supports effective inventory control.

The customer management module provides facilities for storing and managing customer information. Customer details such as names, contact numbers, and email addresses can be recorded and updated whenever necessary. By maintaining organized customer records, businesses can improve communication and strengthen customer relationships.

Similarly, the vendor management module allows businesses to maintain accurate records of suppliers. Vendor information, including names, addresses, and contact details, is stored within the database for easy access. This feature supports efficient procurement processes and facilitates better supplier management.

The order processing module automates customer purchases and transaction management. Whenever an order is placed, the system calculates the total cost automatically and updates inventory levels accordingly. This eliminates the need for manual calculations and significantly reduces the risk of errors. The system also allows users to monitor order status and track deliveries effectively.

The payment management module records financial transactions and supports multiple payment methods such as cash, mobile money, and bank transfers. Businesses can easily determine which payments have been completed and which remain outstanding. This contributes to improved financial accountability and cash flow management.

In addition, the expense management module enables organizations to record and categorize operational expenses. By maintaining accurate expense records, businesses can better understand their expenditure patterns and identify opportunities for cost reduction.

One of the most valuable features of the system is its reporting capability. The reporting module automatically generates sales reports, expense reports, and profit reports based on information stored within the database. These reports provide management with meaningful insights into business performance and support evidence-based decision-making.

Overall, the Smart Business Management System serves as a comprehensive business automation platform that integrates multiple business functions into a single application. Its implementation improves efficiency, enhances data accuracy, reduces administrative workload, and supports sustainable business growth.

Privacy & Legal Compliance

Compliance with Sierra Leonean Legal Frameworks

Our application handles data with strict adherence to regional regulatory frameworks governing information security and digital privacy in Sierra Leone:

Data Protection Policy Framework: In accordance with the Sierra Leone Data Protection and Right to Access Information Policy Framework, our platform implements Data Minimization. The system only requests and processes user inputs that are functionally necessary to compute business outputs (such as entering transaction metrics in customers.py or tracking outlays in expenses.py).

Constitutional Privacy Rights: To respect the individual right to privacy outlined in Section 22 of the Constitution of Sierra Leone 1991, our software architecture guarantees that transient session variables are securely managed. No arbitrary tracking data or personal citizen metrics are recorded behind the scenes without direct user action.

Technical Privacy Safeguards in Code

To ensure compliance at the code level, our group implemented several software engineering best practices within our repository:

Secure Database Storage: User data is structured locally through a relational schema inside business.db. Access is strictly brokered by our dedicated backend script (database.py) to prevent cross-contamination or unauthorized access by outside processes.

No Hardcoded Credentials: In alignment with public open-source repository standards, no private administrative passwords, encryption keys, or unhashed mockup data are hardcoded directly into the Python source files.

Data Volatility: Metrics initialized during active application runtime (e.g., using dashboard.py) remain transient within RAM memory space and are safely cleared upon execution termination, mitigating the risk of persistent data exposure on shared hardware.

Open-Source Licensing Compliance

To ensure complete legal transparency, this project has been published under the MIT License on GitHub. This public licensing model explicitly grants permissions for open-source modification and community distribution while legally indemnifying the original student development group from system misuse or unexpected deployment liabilities.

Structured Programming Application

The development of the Smart Business Management System demonstrates the practical application of structured programming principles. Structured programming is a programming methodology that emphasizes logical program design through the use of clearly defined control structures, modular development, and systematic problem-solving techniques. The objective of this approach is to improve program readability, maintainability, reliability, and efficiency.

The system was developed using Python, which supports structured programming concepts extensively. Throughout the application, variables are used to store and manipulate data associated with products, customers, vendors, orders, payments, and expenses. Variables enable the system to process information dynamically and perform calculations required for business operations. Examples include storing product prices, quantities, customer names, payment amounts, and expense values.

Different data types are utilized within the application to represent various forms of information. Integer data types are used for identifiers and quantities, floating-point values are used for prices and financial calculations, strings are used for textual information such as names and addresses, while Boolean values support logical operations and decision-making processes. The proper use of data types contributes to efficient memory utilization and accurate data processing.

Decision-making structures play a critical role in controlling the flow of execution within the system. Conditional statements such as if, elif, and else are used to validate user input, verify stock availability, authenticate login credentials, and determine payment statuses. These control structures enable the system to respond appropriately to different operational conditions and ensure that only valid transactions are processed.

Iteration structures are also widely applied within the system. Loops are used when retrieving records from the database, displaying information in tables, processing lists of products, and generating reports. The use of loops eliminates repetitive coding and improves program efficiency by enabling multiple records to be processed automatically.

Another important feature of structured programming demonstrated in this project is modularization through user-defined functions. The application is organized into numerous functions, each responsible for performing a specific task. Examples include functions for adding products, managing customers, recording payments, generating receipts, processing orders, and producing reports. By dividing the program into smaller manageable units, the system becomes easier to understand, test, debug, and maintain.

The project also incorporates database interaction procedures that follow structured programming practices. SQL commands are organized within dedicated functions to handle data insertion, retrieval, updating, and deletion. This separation of responsibilities improves code organization and enhances system reliability.

Error handling mechanisms have also been implemented to improve system robustness. Input validation checks ensure that users enter accurate information before records are saved into the database. This minimizes the occurrence of invalid data and helps maintain database integrity.

The application further demonstrates the principle of sequential execution, where tasks are performed in a logical order. For example, a user must successfully log into the system before accessing operational modules. Similarly, products must exist in inventory before orders can be processed. Such logical sequencing ensures consistency and supports smooth business operations.

Through the implementation of variables, data types, conditional statements, loops, functions, database operations, input validation, and modular program design, the Smart Business Management System effectively demonstrates the practical application of structured programming concepts in solving real-world business management challenges.

SDG Relevance

The Smart Business Management System is strongly aligned with the United Nations Sustainable Development Goal 8, which focuses on promoting sustained, inclusive, and sustainable economic growth, productive employment, and decent work for all. The achievement of this goal requires organizations and businesses to adopt efficient systems that improve productivity, enhance operational performance, and support long-term sustainability.

Small and Medium-Sized Enterprises are widely recognized as important drivers of economic growth and employment generation. However, many SMEs face operational challenges that limit their productivity and competitiveness. Inefficient record management, poor financial monitoring, inaccurate inventory control, and inadequate reporting mechanisms often hinder business expansion and profitability. The Smart Business Management System addresses these challenges by introducing a digital solution that improves business management practices and promotes operational excellence.

The system contributes to increased productivity by automating routine administrative tasks that would otherwise require significant manual effort. Activities such as inventory tracking, order processing, payment recording, and report generation can be completed quickly and accurately. As a result, employees can dedicate more time to productive activities that contribute directly to business growth.

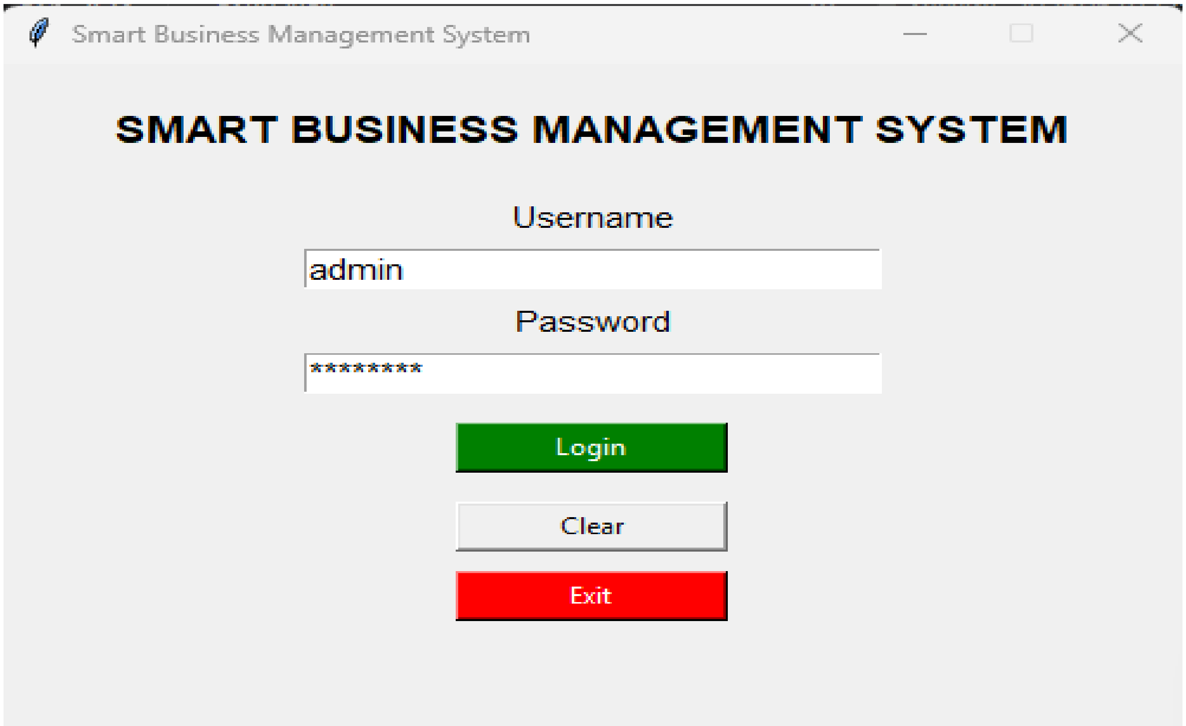
The system also supports financial sustainability by providing accurate records of sales, expenses, and profits. Through automated calculations and reporting tools, business owners gain access to reliable information that supports informed decision-making. Better financial management contributes to increased profitability, improved resource allocation, and enhanced business stability.

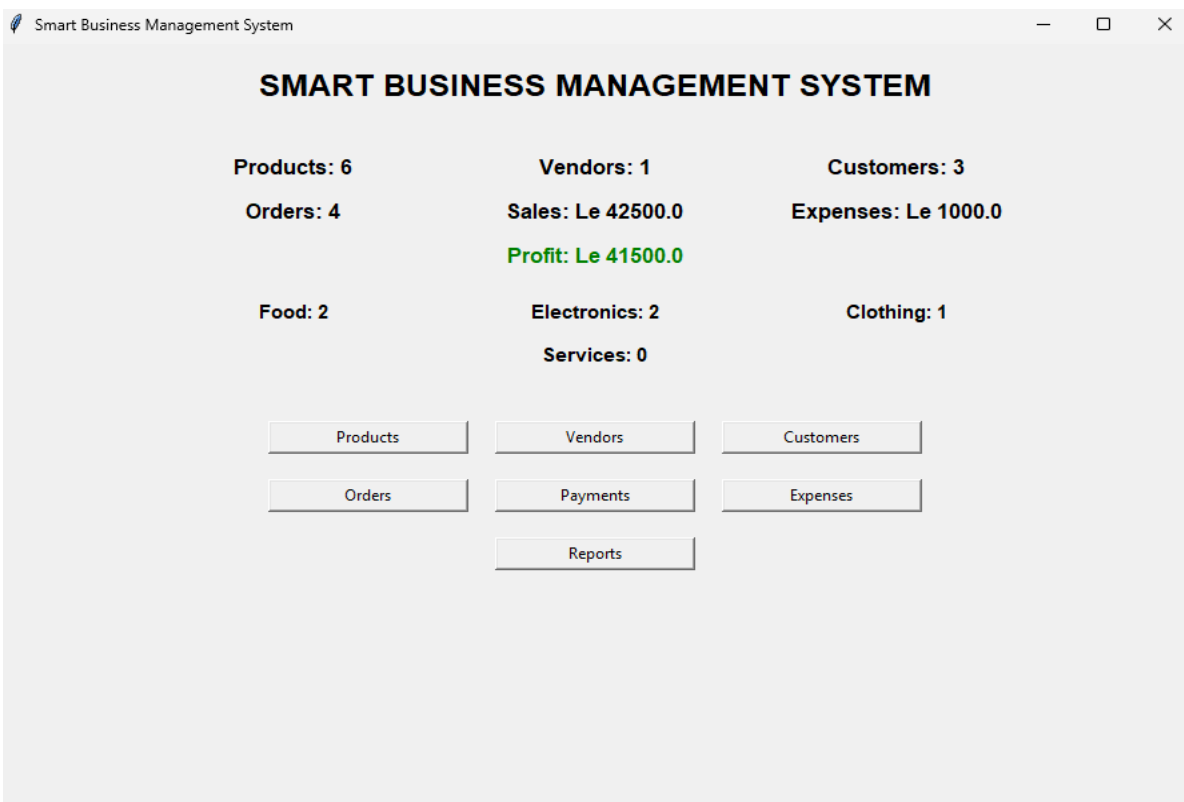
Another important contribution of the system is the promotion of innovation and technological adoption within the business sector. By encouraging businesses to transition from manual processes to computerized systems, the project supports digital transformation initiatives and prepares organizations for participation in an increasingly technology-driven economy.

Furthermore, the system strengthens entrepreneurship by providing business owners with practical tools for managing operations effectively. Entrepreneurs can monitor performance, identify trends, evaluate profitability, and implement strategies for expansion based on accurate data. This capability enhances business competitiveness and supports sustainable economic growth.

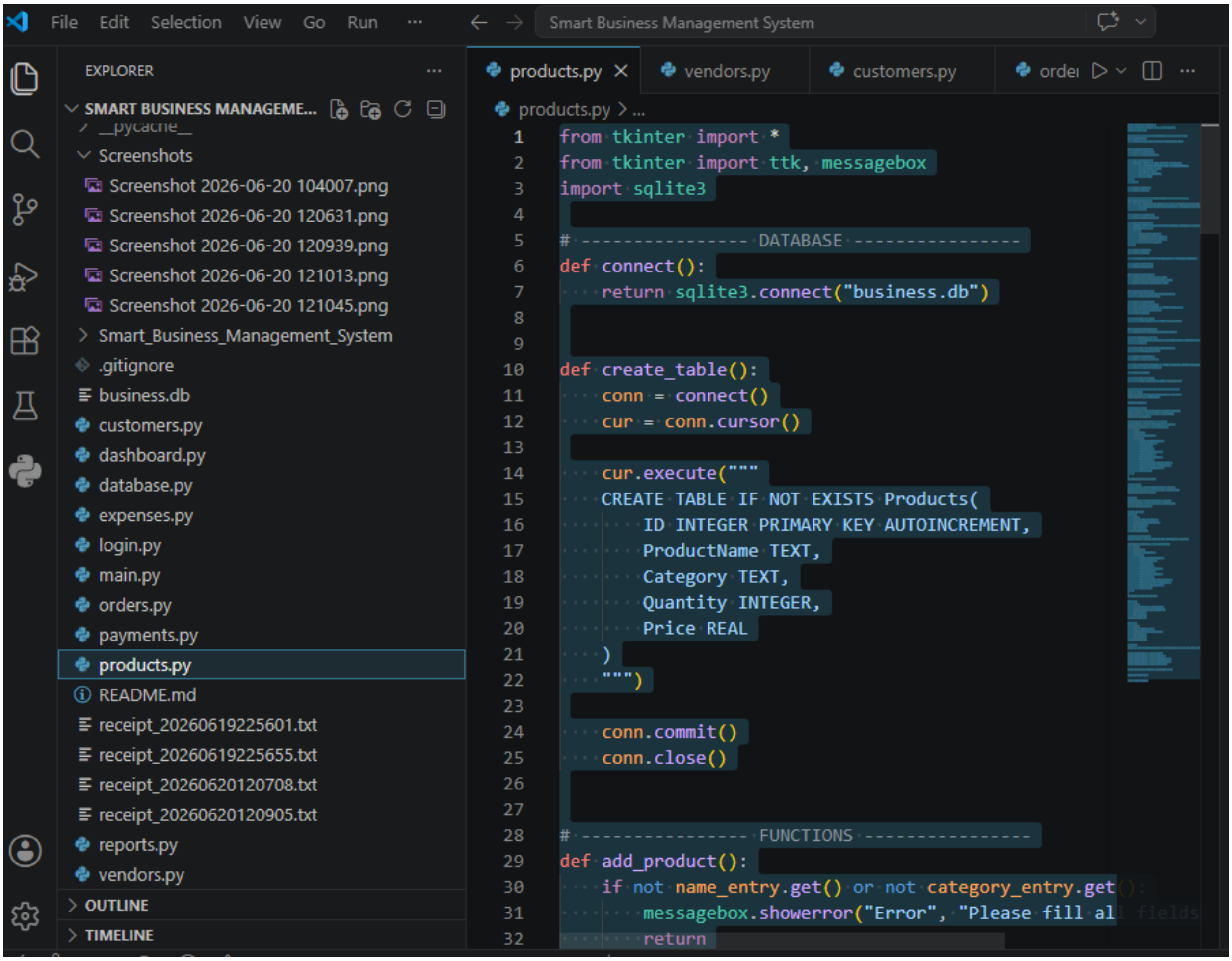
The Smart Business Management System also contributes indirectly to employment creation. As businesses become more efficient and profitable, they are better positioned to expand their operations, create new job opportunities, and contribute to local economic development. Improved business performance can lead to increased investment, higher revenues, and greater economic activity within communities.

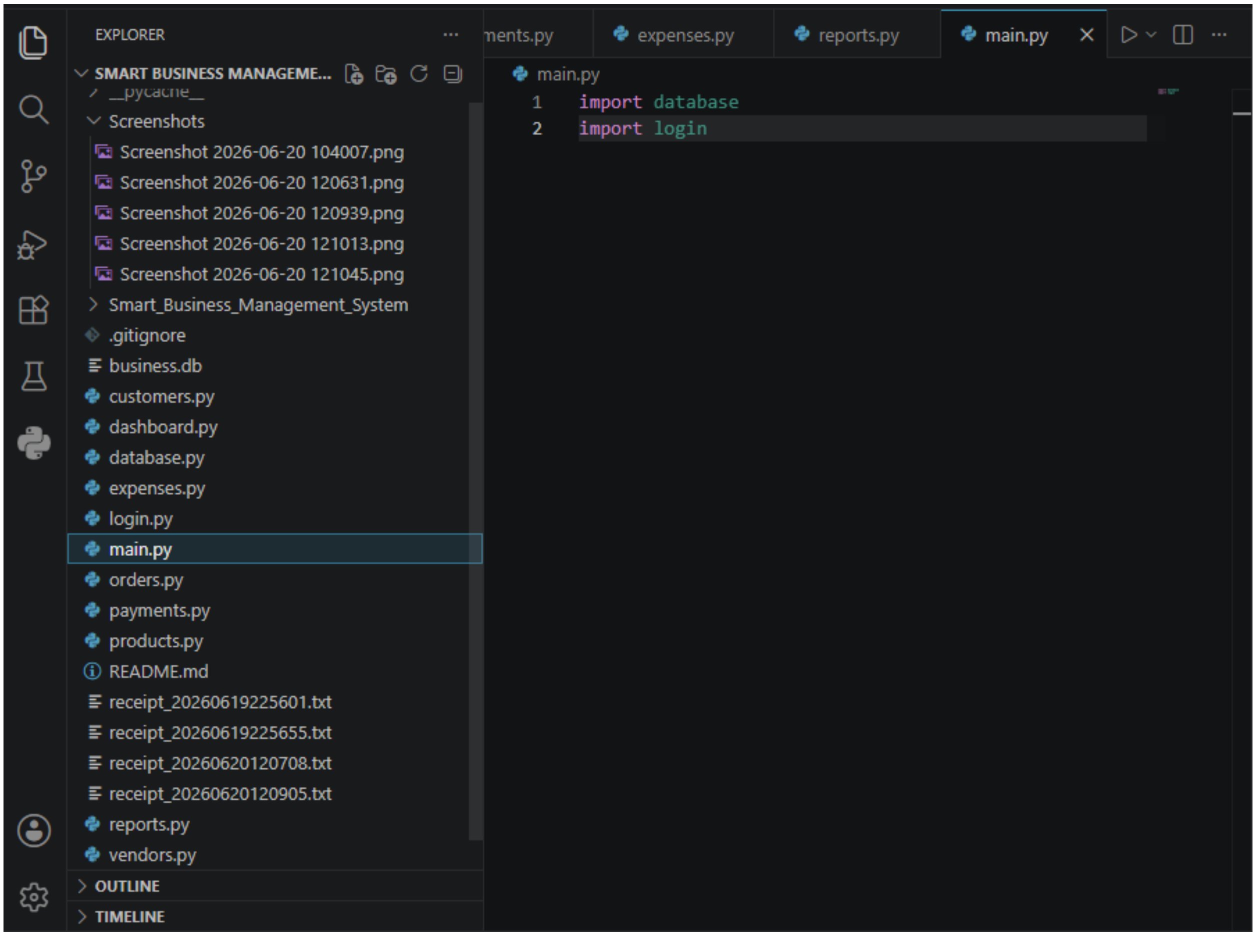
System Screenshots





Source code Screenshots




EXPLORER
SMART BUSINESS MANAGEME...
__pycache__
Screenshots
Screenshot 2026-06-20 104007.png
Screenshot 2026-06-20 120631.png
Screenshot 2026-06-20 120939.png
Screenshot 2026-06-20 121013.png
Screenshot 2026-06-20 121045.png
Smart_Business_Management_System
.gitignore
business.db
customers.py
dashboard.py
database.py
expenses.py
login.py
main.py
orders.py
payments.py
products.py
README.md
receipt_20260619225601.txt
receipt_20260619225655.txt
receipt_20260620120708.txt
receipt_20260620120905.txt
reports.py
vendors.py
OUTLINE
TIMELINE
ments.py
expenses.py
reports.py
main.py
main.py
import database
import login

CONCLUSION

The Smart Business Management System (SBMS) successfully provides a comprehensive solution for managing essential business operations within small and medium-sized enterprises. The project was developed to address the limitations associated with manual record-keeping systems, including data inaccuracies, inefficient inventory management, poor financial tracking, and difficulties in generating business reports. Through the implementation of an integrated digital platform, the system enables organizations to manage products, customers, vendors, orders, payments, and expenses more effectively.

The use of Python, Tkinter, and SQLite has resulted in a reliable, user-friendly, and cost-effective application capable of automating routine business processes. The centralized database structure ensures secure storage and easy retrieval of information, while the reporting features provide valuable insights into business performance. Automated calculations and inventory monitoring functionalities further improve accuracy and operational efficiency.

The project also demonstrates the practical application of structured programming concepts such as variables, data types, decision structures, loops, functions, and database operations. These concepts contribute to the development of a well-organized, maintainable, and efficient software solution capable of addressing real business challenges.

Furthermore, the Smart Business Management System contributes to Sustainable Development Goal 8 (Decent Work and Economic Growth) by promoting business productivity, encouraging digital transformation, and supporting sustainable economic development. By improving business management practices and facilitating informed decision-making, the system helps organizations achieve greater efficiency, profitability, and long-term sustainability.

In conclusion, the Smart Business Management System represents an effective technological solution that enhances business operations, improves record management, supports organizational growth, and demonstrates the importance of information technology in modern business environments.

REFERENCES

Agarwal, B. and Lang, J. (2020) Fundamentals of Database Systems. 7th edn. Boston: Pearson.

Beighley, L. and Morrison, M. (2021) Head First Python. 3rd edn. Sebastopol: O'Reilly Media.

Connolly, T. and Begg, C. (2015) Database Systems: A Practical Approach to Design, Implementation and Management. 6th edn. Harlow: Pearson Education.

Coronel, C. and Morris, S. (2019) Database Systems: Design, Implementation and Management. 13th edn. Boston: Cengage Learning.

Downey, A. (2016) Think Python: How to Think Like a Computer Scientist. 2nd edn. Sebastopol: O'Reilly Media.

Elmasri, R. and Navathe, S. (2017) Fundamentals of Database Systems. 7th edn. Boston: Pearson.

Forouzan, B.A. (2018) Introduction to Computer Science and Programming. New York: McGraw-Hill Education.

Guttag, J.V. (2021) Introduction to Computation and Programming Using Python. 3rd edn. Cambridge, MA: MIT Press.

Lutz, M. (2013) Learning Python. 5th edn. Sebastopol: O'Reilly Media.

Martin, R.C. (2018) Clean Architecture: A Craftsman's Guide to Software Structure and Design. Boston: Prentice Hall.

McKinney, W. (2022) Python for Data Analysis. 3rd edn. Sebastopol: O'Reilly Media.

Pressman, R.S. and Maxim, B.R. (2020) Software Engineering: A Practitioner's Approach. 9th edn. New York: McGraw-Hill Education.

Python Software Foundation (2026) Python Documentation. Available at: https://docs.python.org/3/ (Accessed: 21 June 2026).